23/24 Sem 1 Q1(c)

$$\int \sin(3\theta)\,\underline{\cos^4\theta}\,d\theta \qquad \left[\text{Hint: } \cos 2\theta = 2\cos^2\theta - 1\right]$$

Method 2

$$\cos^4\theta = (\cos^2\theta)^2$$

$$= \left(\frac{\cos 2\theta + 1}{2}\right)^2$$

$$= \frac{1}{4}\left(\underline{\cos^2 2\theta} + 2\cos 2\theta + 1\right)$$

$$= \frac{1}{4}\left(\frac{\cos 4\theta + 1}{2} + 2\cos 2\theta + 1\right)$$

$$= \cdots$$

$$= \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$$

$\therefore \int \sin(3\theta)\cos^4\theta \, d\theta$

$= \frac{1}{8}\int \sin 3\theta \cos 4\theta \, d\theta + \frac{1}{2}\int \sin 3\theta \cos 2\theta \, d\theta$

$+ \frac{3}{8}\int \sin(3\theta) \, d\theta$

[Proceed with Product-to-Sum Formula]

Method 2

$\underset{\substack{\downarrow \\ 2\theta+\theta}}{\sin 3\theta} = \underline{\sin 2\theta} \cos\theta + \underline{\cos 2\theta} \sin\theta$

$= \underline{2\sin\theta\cos\theta}\cos\theta + (\underline{2\cos^2\theta - 1})\sin\theta$

$= \underline{2\sin\theta\cos^2\theta} + \underline{2\cos^2\theta\sin\theta} - \sin\theta$

$= 4\sin\theta\cos^2\theta - \sin\theta$

$= \sin\theta(4\cos^2\theta - 1)$

$$\therefore \int \sin 3\theta \cos^4\theta \, d\theta = \int \sin\theta\,(4\cos^2\theta - 1)\cos^4\theta \, d\theta$$

$$= \int \sin\theta\,(4\cos^6\theta - \cos^4\theta)\, d\theta$$

[Proceed with $u = \cos\theta$]

↳ eliminate $\sin\theta$